Manuscript outline

Cell enlargement, polarity competition and migration persistence

Vision-aligned manuscript outline with proposed main figures and detailed legends

20 October 2026 · Completed Figures 2–4 revisions with stated scientific qualifications

Scientific argument and manuscript plan

Working hypothesis: WGD-associated enlargement changes how established polarity is organized and replaced; these changes may provide a physical basis for greater migration persistence. The theoretical contribution tests size under unchanged local kinetic parameters. Length differs between model cells to represent WGD-associated enlargement, rather than elongation during a turn. Existing migration observations supply empirical motivation; new paired WGD motility results retain a substantial, explicitly reserved contribution.

Figures 2–4 test how challenge shape and timing separate response mechanisms, how physical separation affects competition at fixed size, and how replicated fluctuating input changes polarity-direction persistence and spatial organization. Figure 2 reports equal-area widths and pulse trains. Figure 3 distinguishes cue-driven handover, cue-free resolution and directly measured activity direction. Figure 4 follows its completed hypothesis, coverage and sensitivity decisions. Detailed native histories and every finite-horizon or numerical qualification remain in the linked supplements.

This is a working manuscript outline. Figures 1 and 5 and the reserved Figure 6 experimental section retain their preceding content. Activity polarity, modelled replacement, observed migration and conditional OU velocity memory remain distinct readouts; their biological connection is an experimental hypothesis.

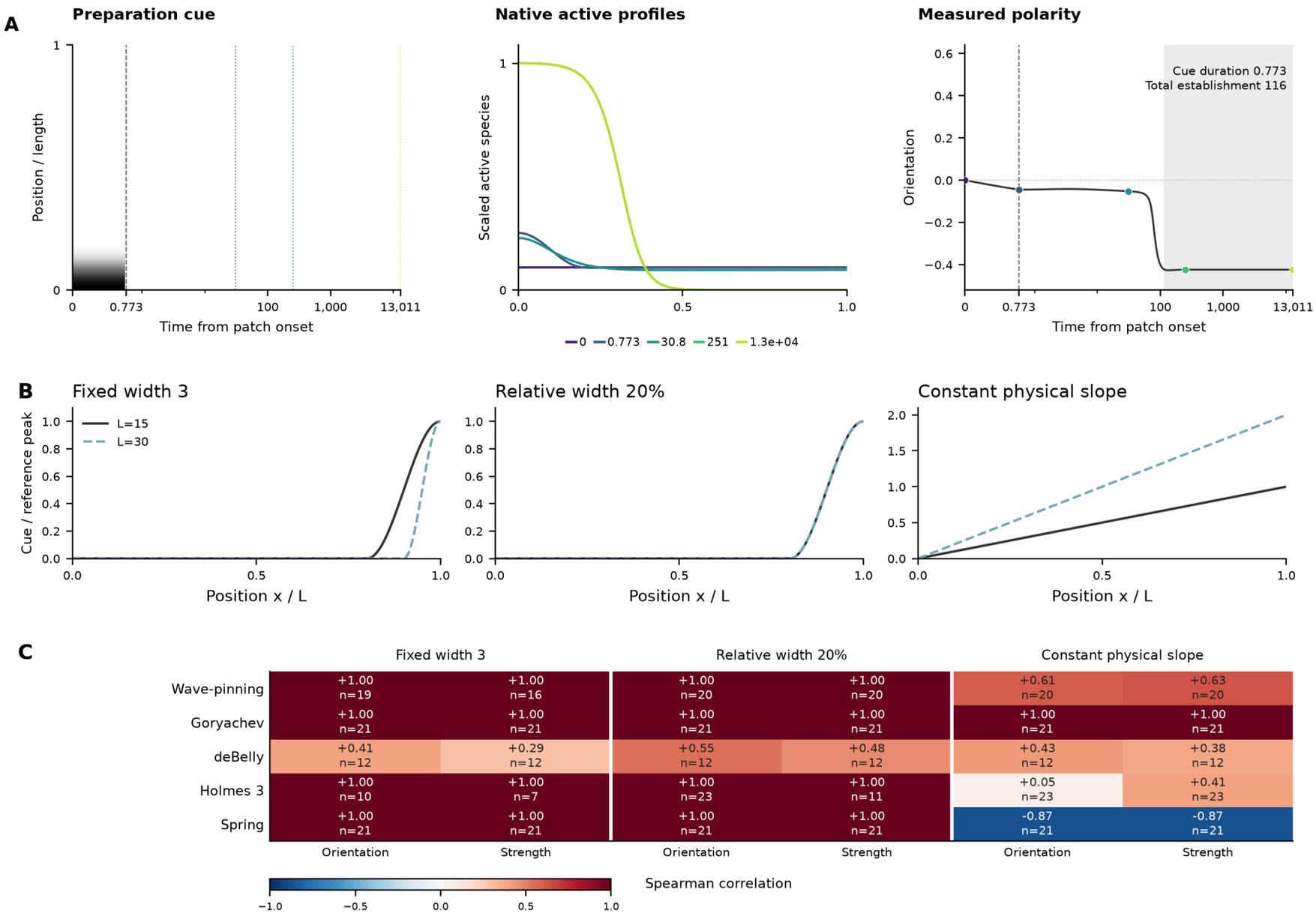
Proposed main figures

- Figure 1. Enlargement changes the cue duration required for lasting polarity replacement
- Figure 2. Shape and timing distinguish replacement responses
- Figure 3. Separation affects native competition differently across models
- A size advantage in heading persistence can coexist with multiple active regions.

Manuscript outline · continued

- Figure 5. Published migration summaries support a persistence hypothesis with conditional OU interpretation
- Figure 6. RESERVED: new WGD motility-assay results

Figure 1. Five models and three finite challenges



Current canonical comparison, 10 October Spearman revision; corrected independent-endpoint deBelly.

Figure 1. Enlargement changes the cue duration required for lasting polarity replacement

Comparative theoretical foundation: five established models, three finite challenges, and two distinct scalar replacement gates.

Detailed legend

A, Example wave-pinning preparation from a homogeneous state using a finite left patch. The cue ends at 0.773 native time units, whereas the subsequent basal relaxation establishes the polarized profile later (reported establishment arrival about 116). Coloured profiles and orientation samples distinguish input duration from autonomous development. The grey late interval is verification, not additional stimulation. Profiles are scaled for display.

B, Three rightward finite challenges restart the identical qualifying nominal-amplitude, relative-20% basal checkpoint at each model and length: fixed physical support 3 (Spring 0.2 native length units), relative support 20%, and a linear ramp with constant physical slope. Patch peaks remain fixed within each model; a fixed-slope ramp has a larger peak in a longer domain. Goryachev patches use the independently selected, length-independent multiplier K=64 (peak 6.4), which makes this a response-permissive comparison rather than a calibrated equal biochemical dose across models.

C, Spearman rank correlation between length and the shortest tested passing pulse duration, separately for orientation replacement and strength replacement, across Wave-pinning, Goryachev, corrected deBelly, Holmes 3 and Spring. Each cell reports the correlation and included successful-length count. All audited scalar successes enter, including those failing the separate native-field tail criterion. At least four distinct successful lengths and nonconstant durations are required; censored and preparation-ineligible cases receive no invented time. The companion duration curves retain all lengths and field-status markers.

After each positive pulse, input returns to the native basal value for 10,000 native time units. Orientation requires all 201 final-500 samples to have scalar orientation above 0.05 and contrast at least 0.05, with orientation range below 0.005. Strength additionally requires every tail orientation to be within 1% of the absolute initial leftward scalar orientation; it does not compare individual pole heights or amounts. The separate native-field tail flag requires pointwise range and projected 500-unit drift within 0.1% of frozen field scales. The reported duration excludes preparation and recovery. It is the shortest passing tested duration, with sampled fail/pass boundaries refined to 1%; narrower unobserved intervals are not excluded.

deBelly uses independently commanded corrected endpoints, rather than the deprecated in-loop centered extension. Spring length and time retain native units and are not silently converted to micrometres or seconds. Model-specific axes and stimulus definitions prevent interpreting cross-model amplitudes or clocks as biologically calibrated equivalents.

Figure 1. Enlargement changes the cue duration required for lasting polarity replacement · continued

Narrative points and conclusions

- Fixed-width and relative-width patches show perfectly increasing rank order among observed successes in Wave-pinning, Goryachev, Holmes 3 and Spring; enlargement often increases the finite stimulation required for a retained response.
- The response depends on the cue geometry and model. Spring has a negative constant-slope association (Spearman correlation -0.874 for both gates), while Holmes orientation under the constant-slope cue is nearly unassociated with length (0.047).
- Corrected deBelly curves are nonmonotone despite moderate positive rank correlations. The full duration curves and eligibility ranges are essential to the result.
- These are deterministic polarity-replacement experiments. They motivate, but do not measure, spontaneous polarity memory or migration persistence.

Proposed measurements and experimental follow-up

Mechanism-linked cellular follow-up would match local input support, amplitude and duration, track the original and challenged activity regions through stimulation and withdrawal, and distinguish orientation replacement from recovered activity strength. This validation is proposed and has no experimental results in this outline.

Scope, limitations and pending validation

Correlations are conditional on observed scalar success, rather than an unbiased summary of all possible sizes or a universal scaling law. Ineligible starts are distinct from responses unobserved by the finite horizon.

A retained scalar response can coexist with a changing native field. Only 13 of the 72 corrected deBelly scalar successes pass the separate native-field tail flag. Resolution checks from the earlier centered-deBelly campaign cannot qualify the corrected replacement thresholds.

Sources and reproducibility

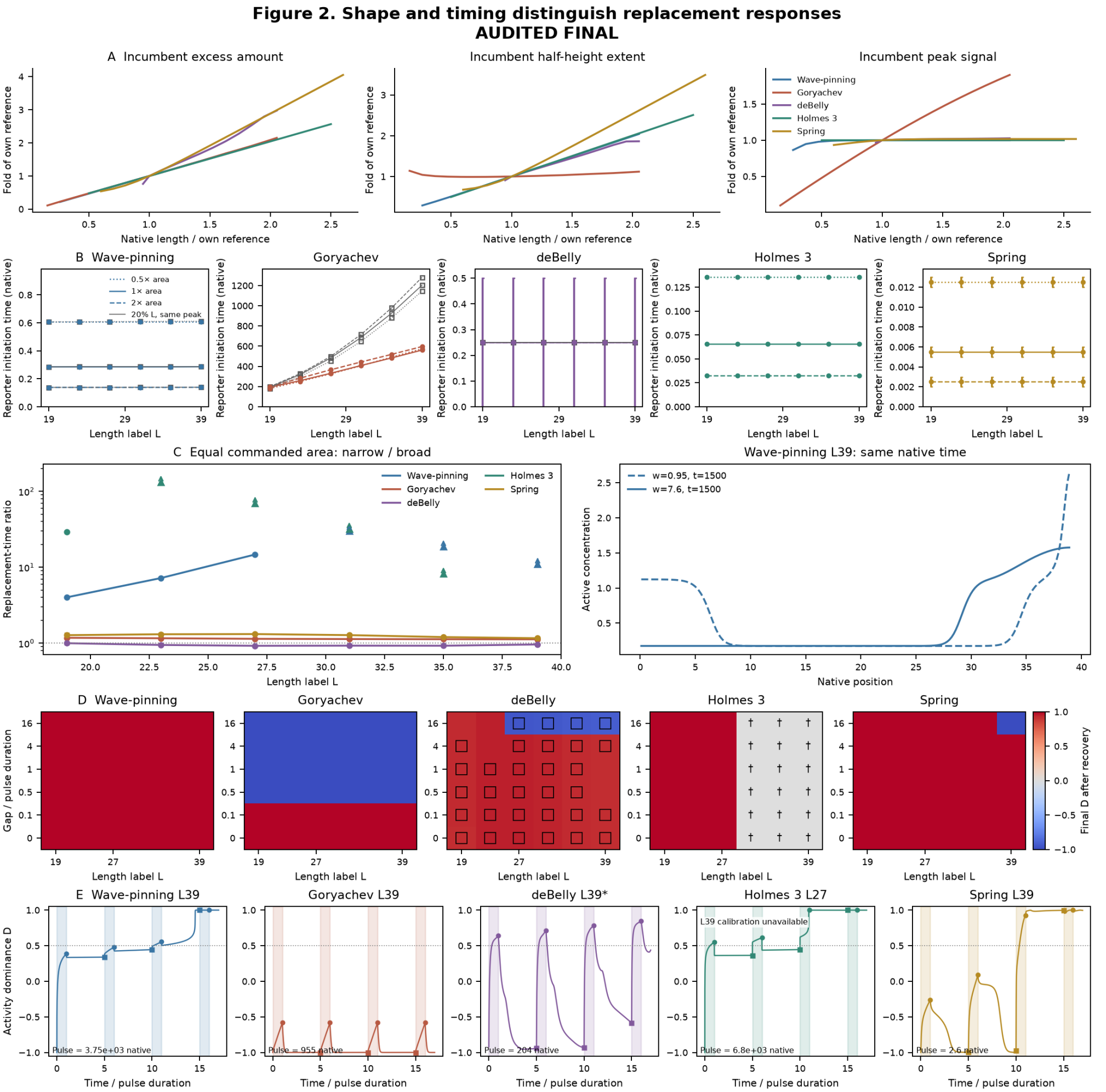
Three-test protocol and Spearman rule

Correlation table

Corrected endpoint method

Current duration curves

Figure 2. Shape and timing distinguish replacement responses



Model	Trace L	Initiation L39/L19	Narrow / broad, L19 → L39	Pulse 1 → gap 1: D	Final D
Wave-pinning	L39	1–1.01	3.99–4.05 → >11.1	0.39 → 0.34	1.00
Goryachev	L39	2.87–2.92	1.17–1.18 → 1.12–1.13	-0.58 → -1.00	-1.00
deBelly*	L39	both in (0, 0.5]	0.962–1.04 → 0.955–0.97	0.64 → -0.94	0.92
Holmes 3	L27	0.985–1.02	29–29.1 → uninformative	0.55 → 0.36	1.00
Spring	L39	0.833–1.2	1.2–1.36 → 1.15–1.18	-0.26 → -1.00	1.00

Shape: beam width at fixed input area • Timing: equal total ON pulse trains • Reporter: activity share after each gap

↑ triangles: finite-cap ratio lower bounds; bounds cannot be ranked. †: no calibration; ×: low activity; □: failed field tail.

* deBelly later-gap timing and endpoint fields have numerical qualifications (S5). Native clocks differ; optical dose is uncalibrated.

Five-model shape and timing signature assays

Figure 2. Shape and timing distinguish replacement responses

Test whether shape and timing of a measured challenge distinguish explanations for size-dependent replacement resistance.

Detailed legend

A, Retained organization of all 97 qualified basal states. Excess amount is integrated inside the interpolated basal half-height core. Amount, extent and native peak signal are divided by each model's own reference values; Spring retains its physical domain and native resting-length convention.

B, Reporter initiation: first cue-site positive-excess peak reaching 10% of the exact L19 basal excess peak, held fixed within each model across all lengths, widths and strengths. The fixed physical-width raised-cosine cue has support 3.8 chemical units or 0.1 Spring units. Dotted, solid and dashed coloured curves have 0.5, 1 and 2 times the reference commanded spatial input area. Grey open-square curves retain the same peak at relative support 20% of length for the three focal models; this diagnostic changes spatial area with length. Bars are saved-sample crossing brackets. deBelly events in the first native update remain in the interval (0, 0.5]; substep interpolation is descriptive only. Open diamonds at zero denote a footprint already above the reporter threshold before cue application. Native clocks are shown separately.

C, Time to eligible D ≥ 0.5 during continuous cue ON for width 0.95 divided by width 7.6 (Spring widths 0.025 and 0.2), at the central fixed commanded spatial area equal to native reference peak times 3.8 (Spring times 0.1). Bars propagate both sampled crossing brackets; upward triangles and arrows are strict finite-cap lower bounds when the narrow arm remains below replacement at its native horizon. Censored bounds are not finite replacement times and cannot be ranked as measured ratios. The profile inset compares exact native L39 fields at the predeclared saved time nearest 1500 with one common concentration scale.

D, Final activity dominance after four equal pulses with total ON 1.5 times each model–length's postwithdrawal commitment calibration, then 10,000 native cue-free units (4000 for deBelly). Gap durations are multiples of one pulse. E, Four pulses with gaps four times one pulse at L39 where calibrated; otherwise the largest calibrated length is used. This deterministic fallback shows Holmes 3 at L27, explicitly labelled, because L39 has no sampled continuous-cue commitment through 100,000 ON units. Selection depends on calibration availability, not train fate. Shading marks cue ON; circles and squares mark pulse and gap endpoints. Time is divided by each model's actual pulse duration, printed in native units. F, The trace-length column preserves this exception beside the numerical signature summary. Shared responses at calibrated lengths do not establish a matched-length comparison at L39.

Dominance D is the positive active-signal excess in the challenger half minus that in the incumbent half, divided by their sum. The instantaneous profile minimum is the common floor. Eligible D = 0.5 means 75% challenger share; low-activity states require explicit amount and contrast gates and receive no invented D. A square marks a failed finite

Figure 2. Shape and timing distinguish replacement responses · continued

native-field tail independently of final fate. Grey dagger cells have no sampled continuous-cue commitment through the explicitly searched finite cap, so no calibration duration or train trajectory is assigned. Plain grey cells lack a completed independent audit; neither is censoring of an executed train.

Commitment calibration is the shortest sampled continuous ON duration, bracketed by native-duration bisection, that yields eligible D ≥ 0.5 after withdrawal and the declared model-native recovery period. The finite native-field tail is evaluated independently and does not redefine scalar replacement. Native clocks differ between models; no absolute cross-model time ranking is used.

The deBelly trace and summary carry an asterisk because later-gap commitment timing is numerically sensitive. At L39, the third-gap endpoint is D = −0.586 with the native 0.5-unit update and D = +0.190 in the separately qualified 0.25-unit companion. The first two gaps still collapse and final replacement agrees, but later-gap timing and field-tail status are not resolution invariant. The L39 401-cell train also has native endpoint-field sensitivity despite phase-D and fate agreement. S5 retains exact-native-start and separately qualified quarter-step preparations, every endpoint, native-field errors and all gate outcomes. Noisy deBelly ensembles use the native 0.5-unit update and remain conditional on that numerical scheme.

All five models keep their native kinetics. Commanded spatial input area per ON time is not calibrated photon dose or realized activated molecular amount. A width penalty or gap retention is an observable signature, not a causal allocation to a particular feedback term or unique molecular identification. Holmes 3 and Spring determine whether supported signatures are shared. Polarity replacement is distinct from observed migration and conditional OU persistence estimates.

S3 reports conditional retention through weakly driven gaps, separately from final replacement. At the third gap with gap duration 16 times one pulse, using the predeclared 80% raw-share criterion with D ≥ 0, counts at mean input 0.5% and 2% of native peak are Wave-pinning L19: 11/12 and 8/12; Wave-pinning L29: 12/12 and 12/12; Wave-pinning L39: 12/12 and 12/12; deBelly L19: 12/12 and 11/12; deBelly L29: 0/12 and 0/12; deBelly L39: 0/12 and 0/12. These are independent-seed denominators, with exact 95% intervals shown in S3; the three gaps of one trajectory are not independent replicates. Mean and zero controls remain visible. Weak describes instantaneous bounded input (maximum 1% or 4% of native peak); at L39 with gap multiple 16 and mean input 2%, whole-domain gap exposure is about 2.46 times targeted ON exposure. This commanded native-coordinate coefficient is not photon dose or realized activation. deBelly retention remains conditional on the native 0.5-unit update.

Narrative points and conclusions

- Narrow-to-broad replacement-time ratios at L19 and L39 are Wave-pinning: 3.99–4.05 → >11.1; Goryachev: 1.17–1.18 → 1.12–1.13; deBelly*: 0.962–1.04 → 0.955–0.97;

Figure 2. Shape and timing distinguish replacement responses · continued

Holmes 3: 29–29.1 → uninformative; Spring: 1.2–1.36 → 1.15–1.18. These are sampled bounds, not fitted scaling laws.

- Activity-share trajectories and their exact gap endpoints test whether an early challenged state persists through cue-free gaps. A later retained replacement does not establish that early activity itself persisted.

Proposed measurements and experimental follow-up

Vary physical cue support at fixed commanded spatial input area and measure time to eligible75% challenger activity share.

Apply equal-total-ON pulse trains, measure activity share at every pulse and gap endpoint, then report postwithdrawal fate and field stability independently.

Scope, limitations and pending validation

Native-to-optical calibration is absent. Native clocks, reporter species, amount scales and stimulus channels differ across models.

Observable signatures do not uniquely identify a molecular mechanism or allocate causation to one feedback term. All five models determine whether responses are shared.

deBelly early activity collapse and final replacement survive the qualified quarter-step check, but the third-gap dominance sign and finite-field-tail status change; the L39 refined-mesh train also has endpoint-field sensitivity. Later commitment timing and native-step noise probabilities are not claimed to be numerically invariant.

Finite native-field tails qualify only the declared horizon; deterministic activity replacement does not measure migration persistence.

Sources and reproducibility

Independent coverage and scientific source receipts

Sampled width-ratio bounds

Selected pulse/gap endpoint measurements

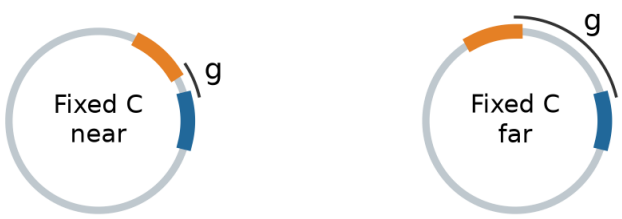
Material deBelly temporal qualification and all phase gates

Complete bounded-noise retention and finite-seed uncertainty

All numerical companions and material qualifications

Figure 3. Separation and native competition · image 1 of 2

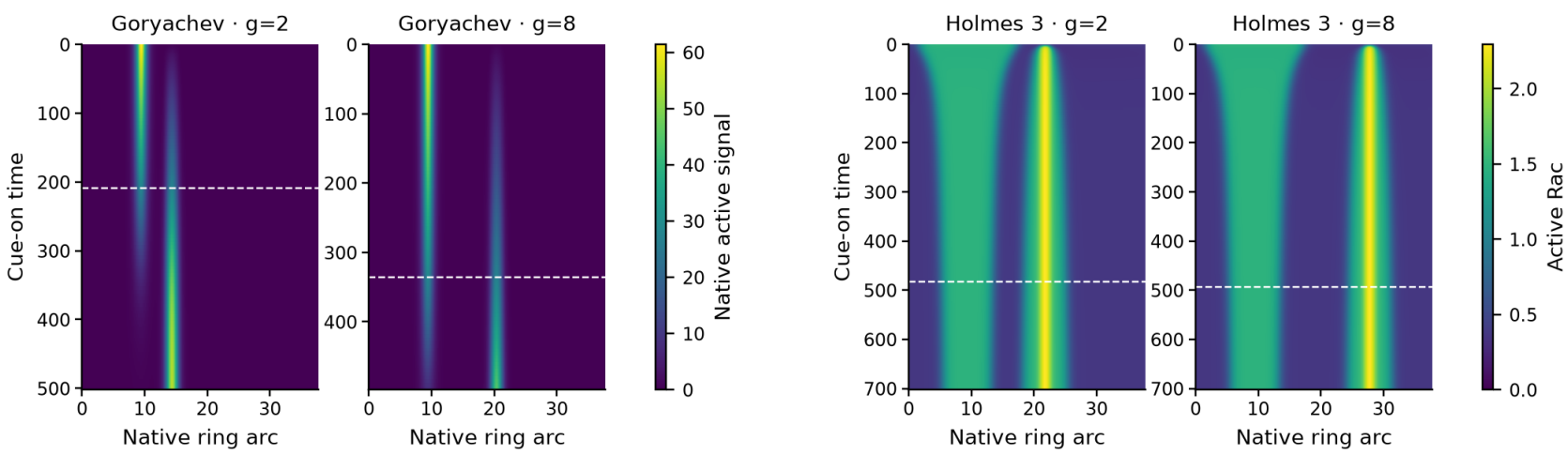
Figure 3 · Separation affects native competition differently across models



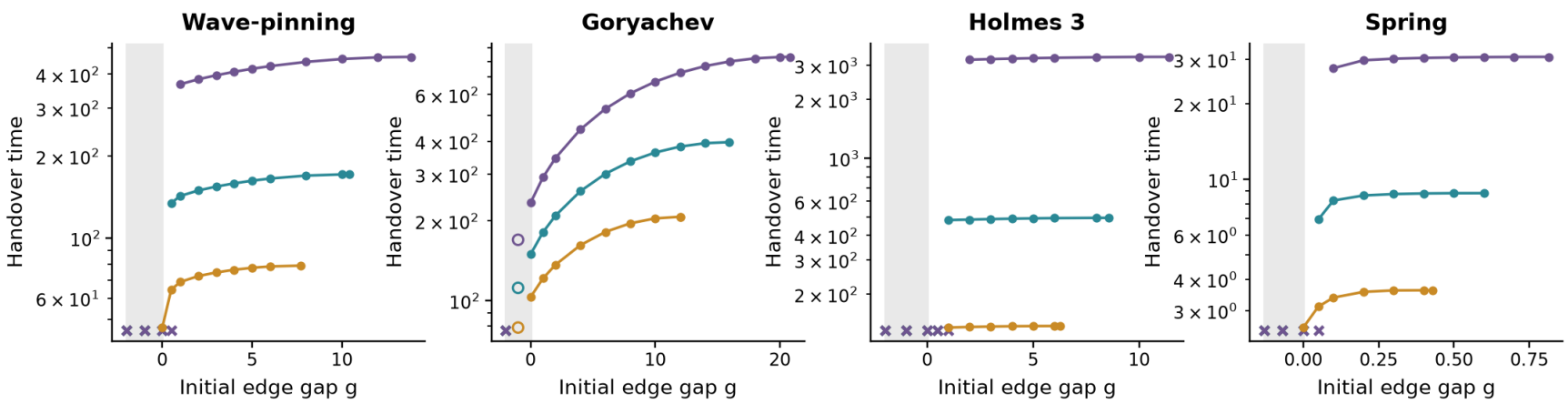
A Rotate one native cue at fixed size

g: initial incumbent 10% edge → cue near edge
A1: continuous cue → complete-region handover
A2: own-handover-scaled pulse → cue-free fate
B: seed ratio 0.5 / 1 / 2 → cue-free peak resolution
12 fresh qualified basals; native kinetics unchanged

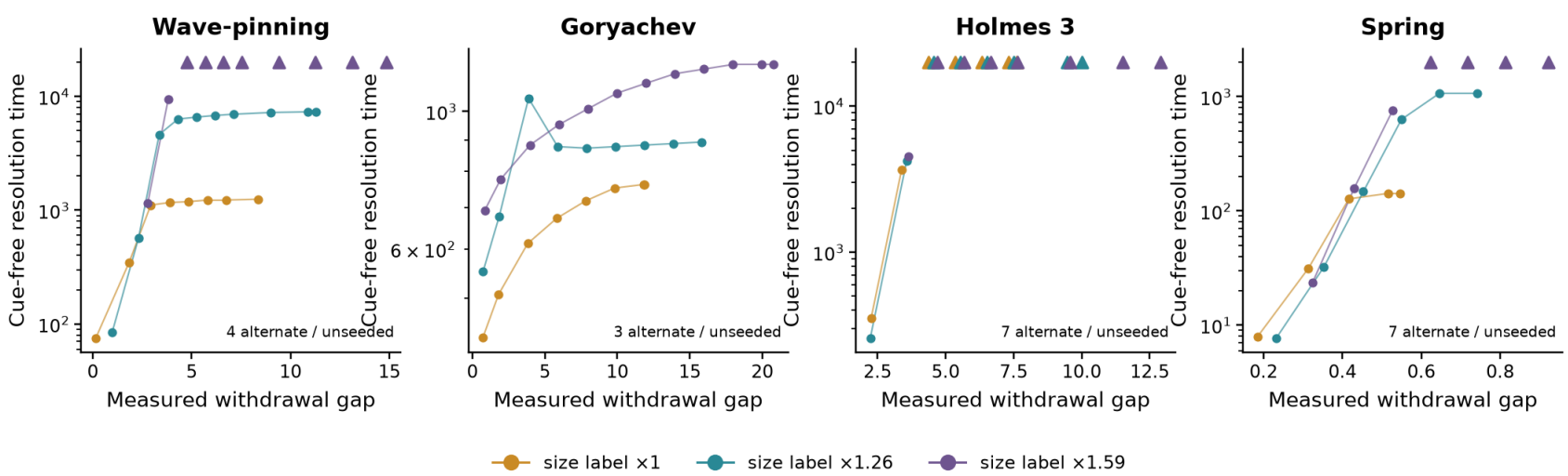
B Same-size native histories · white line: complete-region handover



C1 Driven handover · open symbols: borderline placements

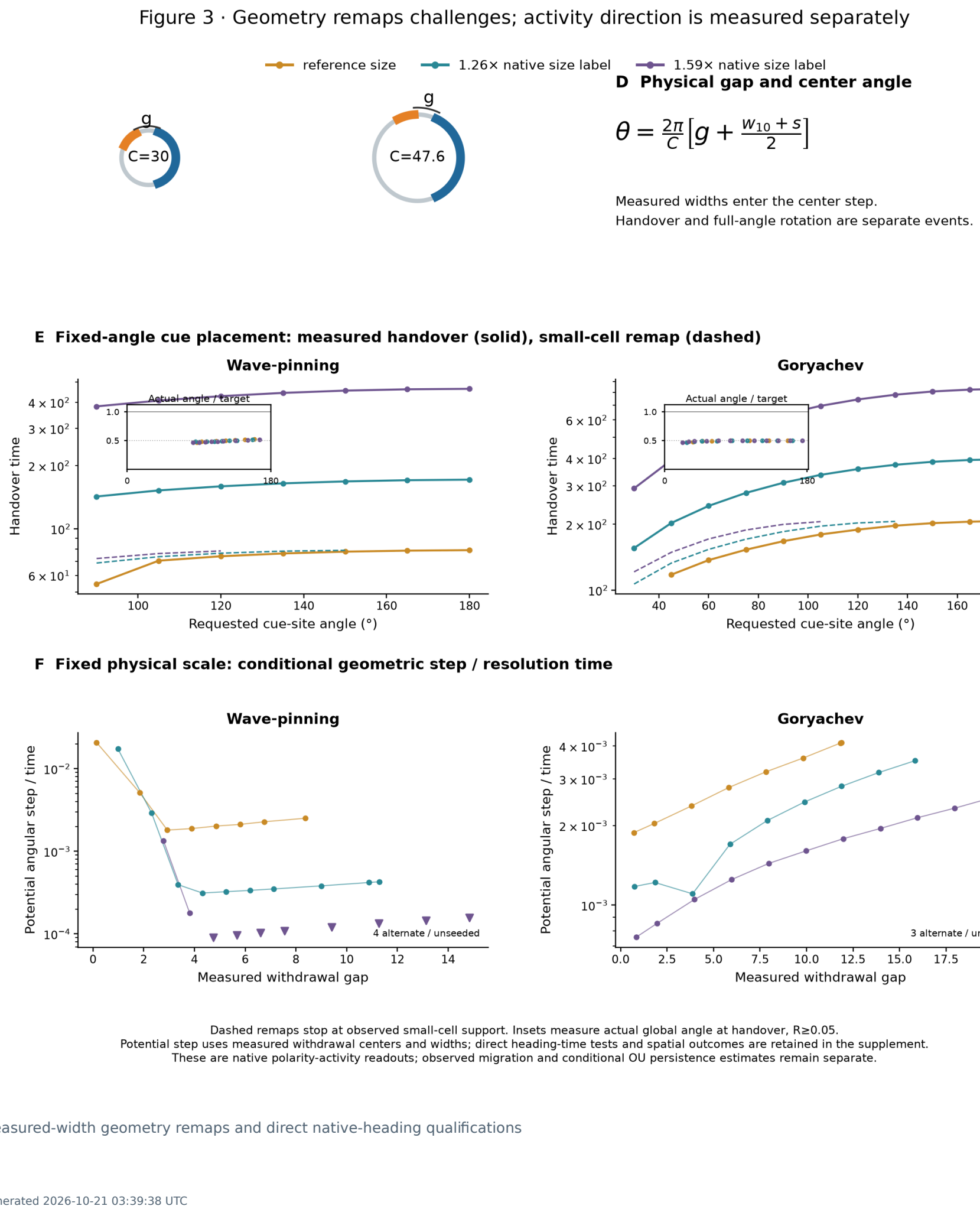


C2 Cue-free resolution · ratio 1 · triangles retain finite-cap lower bounds



Fixed-size cue placement, native histories and separation responses

Figure 3. Geometry and measured activity direction · image 2 of 2



Measured-width geometry remaps and direct native-heading qualifications

Figure 3. Separation affects native competition differently across models

Test whether greater incumbent–challenger separation slows competition at fixed circumference, then examine its geometric translation into measured activity direction.

Detailed legend

A, A native cue is rotated at fixed size. The primary initial gap g runs from the incumbent initial 10%-contrast edge to the cue support near edge along the shorter route. Chemical circumferences are 30, 37.8 and 47.6; Spring resting-size labels are 2.5, 3.15 and 3.97, with actual circumference recorded. All 12 fresh single-region basals qualify without retuning. External placements alone enter the distance claim; borderline negative gaps use open symbols and internal references are supplementary.

B, Native Goryachev and Holmes 3 active-field histories at circumference 37.8 and external gaps 2 and 8. The dashed white line marks independently replayed complete-region handover. Holmes 3 uses active Rac. Chemical A1 histories complete 5, 000 or 20, 000 native units according to the retained continuation rule; Spring completes 2, 000. Every bound uses its actual completed horizon.

C1, Continuous-cue handover is the first time challenger excess active amount exceeds incumbent excess while two resolved, spatially associated regions exist. C2, Cue-free resolution follows seeding to complete-region challenger/incumbent ratio 1; all 0.5/1/2 results are supplementary. Its axis uses the measured current global 10%-contrast gap at withdrawal. Triangles retain completed finite-cap time lower bounds. Unseeded, overlapping-contour, merged and single-region outcomes remain explicit. Lines do not bridge unavailable, censored or topology-changing placements.

D, Requested center angle combines the physical edge gap, half the measured incumbent width and half the cue support, divided by circumference and expressed in radians. The relation is typeset in the main panel. A fixed edge gap is not automatically a fixed center displacement when native incumbent width changes with size. Current B widths, actual circumference, short/long routes and measured center displacement are retained.

E, Solid curves show measured handover time for requested cue angle; dashed curves evaluate the reference-size handover curve at larger-cell measured-width gaps. The geometry-only remap stops within observed reference support and does not interpolate through missing/censored neighbors. The inset directly measures actual global-heading angle at handover, typically about half the requested angle. Handover time is not time to complete the requested angular turn.

F, The fixed-physical-scale view divides measured region-center angular step by measured cue-free resolution time, with censoring propagated. This conditional potential rate is distinct from the directly observed vector-heading change. Full-angle target-time brackets,50/90/99% sensitivities, final headings, coherence coverage and cancellation gaps are supplementary.

Figure 3. Separation affects native competition differently across models · continued

Complete active amounts integrate native active concentration above a common instantaneous minimum over the two full valley-bounded basins, including signal outside the cue. Continuous boundary position uses a local three-point parabolic minimum; actual field amounts use exact piecewise-linear fractional-cell integration. Active flat/nonunique saddles are unscorable. Single moving regions and merged fields do not supply invented two-region handovers or competition times.

The global vector is the first spatial harmonic divided by instantaneous mean active field; $R = |M_1|/M_0$ is coherence, unlike Figure 4 basal-M1-normalized magnitude. Raw exact-direction availability uses $R > 10^{-10}$, with displayed $R \geq 0.05$ and 0.1/0.2 sensitivities. Direction-quality/cancellation gaps break angle unwrapping. All 1, 150 histories include saved-interval principal angular-rate time courses; these are not instantaneous RHS rates or counts of unobserved complete turns.

A2 uses 0.5, 0.75, 1, 1.5 and 2 times each placement own corrected handover, then 13, 000 cue-free chemical units or 1, 300 Spring units. B withdraws at first valid ratios 0.5, 1 or 2, then 20, 000 chemical units or 2, 000 Spring units. Native finite-field stationarity is reported independently of scalar/spatial fate. No finite cap substitutes for an unavailable own handover.

All native positivity and conserved-pool audits pass. All 24 B mirrors agree. A1 handover and B seed-time numerical differences are at most 0.0449%;35/36 B resolution brackets overlap. Goryachev C37.8, gap 4, ratio 1 shifts from 1040–1060 to 1080–1100 units on the finer mesh, about 4% in resolution timing. Native clocks differ; no absolute cross-model time ranking or cell-velocity law is imposed.

Narrative points and conclusions

- Under continuous forcing, Goryachev handover increases with separation, while Holmes 3 is nearly flat over valid two-region placements. Wave-pinning and Spring show near-edge increases followed by broad plateaux. These are model-dependent responses under the declared cue, not a universal absence of distance effects in Wave-pinning.
- Cue-free competition has a different distance response: Wave-pinning, Holmes 3 and Spring can coarsen very slowly, with many large-size resolution lower bounds. Goryachev includes a nonmonotone near-edge response at ratio 1; the full curve and every ratio are retained.
- Actual global-heading measurements qualify the proposed geometry-to-persistence inference. Complete-region handover typically occurs at about half the requested cue angle. Cue-free fields can resolve to an intermediate heading, so a potential center step is not always the realized angular event. These data do not establish an unconditional geometry-driven persistence gain.

Figure 3. Separation affects native competition differently across models · continued

Proposed measurements and experimental follow-up

At fixed native size, vary only cue placement and measure complete-region handover, incumbent 50% excess loss and challenger reaching 50% of basal incumbent excess.

Seed two complete regions to a declared excess ratio, withdraw the cue and measure their actual 10%-contrast separation, first single-peak resolution time, winner association and exact cue-free RHS exchange budget.

Measure the same-run global activity vector, its coherence, actual angular displacement, full-target time brackets and saved-interval angular rates; compare them with geometric remaps.

Scope, limitations and pending validation

Basal width, center displacement and edge gap differ; fixed edge gap is not automatically fixed physical center step across sizes.

A native activity-vector heading is not observed migration direction, a cell velocity law or a conditional OU persistence estimate.

Finite caps produce lower bounds, not proof of permanent arrest. Scalar/spatial winner and full native-field stationarity remain separate.

Two resolved peaks with merged primary 10% contours are ineligible for the primary B gap claim. One moving/merged region is a separate outcome.

The finer-mesh Goryachev B ratio 1 example has about 4% resolution-time sensitivity despite seed-time and winner agreement.

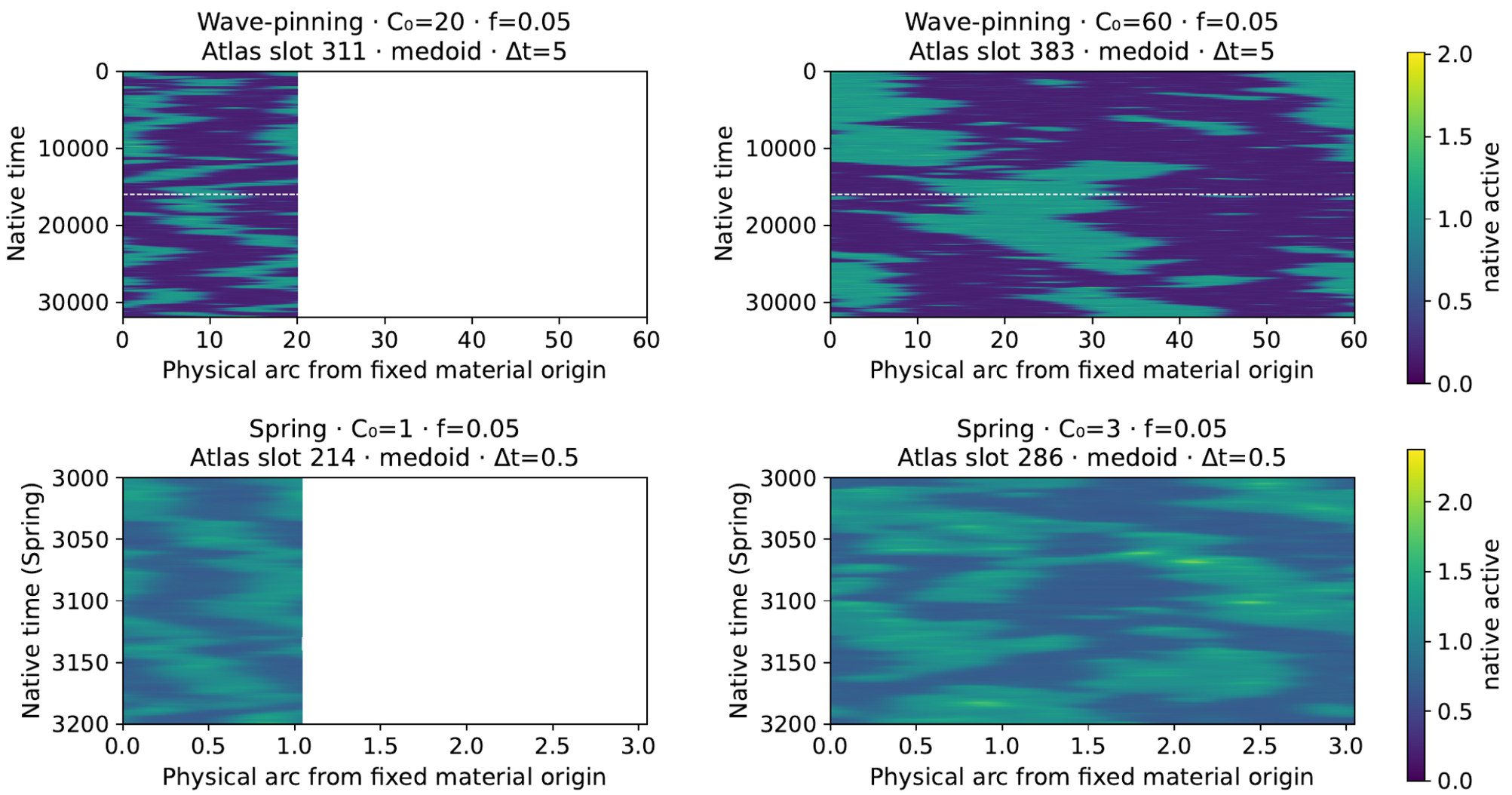
The native-pattern responses do not uniquely identify a molecular mechanism or provide causal allocation to one kinetic term. No noise, moving cue, dose sweep or deBelly ring enters this figure.

Sources and reproducibility

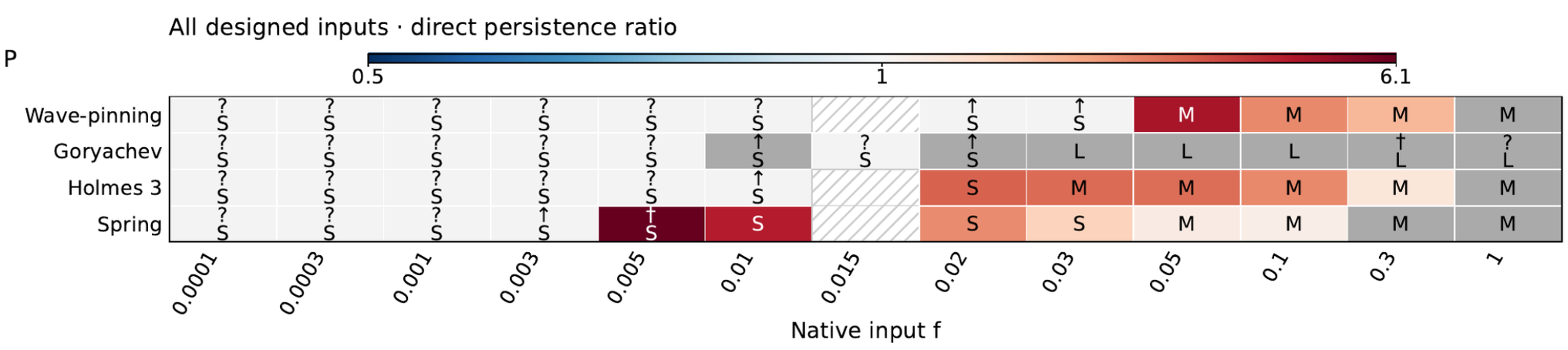
Complete corrected A1 evidence
All A2 switching brackets and unavailable calibrations
All B gap, fate, censoring and actual-heading measurements
Support-limited measured geometry remaps
Direct global-heading target checks
B numerical convergence including the material exception
Final artifacts and source-link audit
Lossless full native-history pixel audit

A size advantage in heading persistence can coexist with multiple active regions.

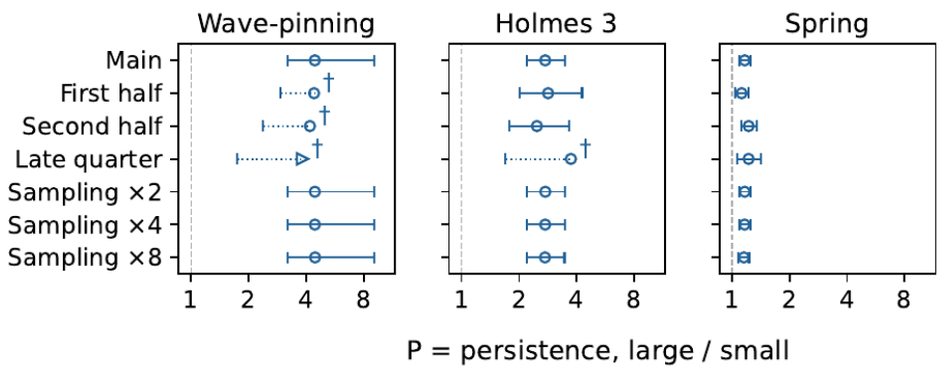
A



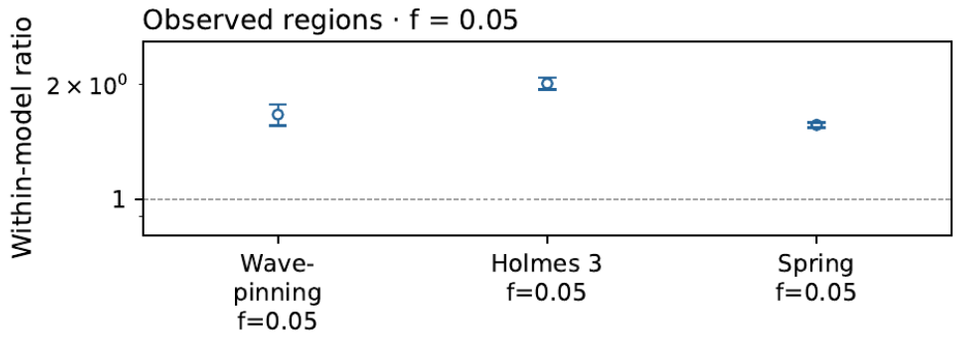
B



C Heading persistence · f = 0.05



D



A size advantage in heading persistence can coexist with multiple active regions.

A size advantage in heading persistence can coexist with multiple active regions.

Observed coexistence of a size advantage in activity-heading persistence with multiple active regions, with full hypothesis and qualification evidence.

Detailed legend

A. Native activity kymographs show the fixed, reviewed f=0.05 medoids for Wave-pinning C_0=20/60 and Spring C_0=1/3. Physical arc, native time and activity colour limits are shared within each model pair. Wave-pinning shows the full 0–32000 native horizon, with a white dashed line at the retained second-half analysis start. Spring shows the declared fixed final one-sixteenth viewport 3000–3200 for both sizes; its full 0–3200 histories remain in S11 and all statistics use the full declared analysis window. Full native grids and exact selected IDs are in the source inventories; concentrations are neither resampled nor quantized. Spring boundary geometry follows its actual circumference.

B. $P=\tau_p(\text{largest})/\tau_p(\text{smallest})$ uses direct heading persistence, with a logarithmic colour scale centred on one. The full grid retains all four models and every designed input. ? is unidentified or unavailable; arrows are one-sided empirical bounds; ↕ is a finite empirical interval; † marks an upper-unbounded bootstrap CI. Bound cells have no fabricated ratio colour. S/M/L/X denote the largest-cell native single-front/multiple-front/low-contrast/mixed regime. Grey retains operational forcing; hatching marks an input not designed for that model.

C. Direct persistence ratios at f=0.05 are shown for all seven declared windows/cadences. Circles are finite ratios; open right-pointing triangles are empirical lower bounds. Horizontal intervals and vertical caps show whole-seed 95% bootstrap uncertainty. A dagger is an unbounded CI upper endpoint; its finite lower endpoint remains distinct from the empirical bound. All three models have a positive lower confidence bound above one in every shown view. Sampling variants reuse the same independent histories.

D. Primary optical region-count quotients compare large/small within each model at f=0.05. The imaging proxy uses fixed physical Gaussian smoothing 1.5 (Spring 0.15), threshold 0.5 and native cadence 80 (Spring 8). Caps retain the exact whole-seed 95% intervals, including Spring’s narrow interval. These counts describe observed local organization and are not a validated predictor of persistence.

The selected f=0.05 conditions were not selected for E4 extension. Their common and longest-arm tables reuse the same common-horizon cohort; agreement between those arms is not independent long-horizon corroboration. Each selected size has 12 independent noisy seeds. Native clocks differ between chemical models and Spring; no cross-model absolute time or amplitude ranking is implied.

Figure 4 R is the native first-mode magnitude divided by its exact no-input basal first-mode magnitude. It can exceed one and differs from Figure 3 coherence |M1|/M0. Global-heading-derived physical-mobility proxies are distinct from qualified single-region literal site MSDs.

A size advantage in heading persistence can coexist with multiple active regions. · continued

Narrative points and conclusions

- At f=0.05, direct large/small heading-persistence ratios are 4.44 for Wave-pinning,2.74 for Holmes 3 and 1.17 for Spring. All seven declared views retain a positive lower confidence bound above one. The largest-size native multiple-front fractions are 0.639,0.857 and 0.659 respectively.
- The complete predeclared hypotheses remain qualified: H1 is inconclusive, H2 does not meet its overall criterion and H3 is inconclusive in both primary observation arms. The full S0 memo shows every model, deciding numerical shortfall and two alternative framings. A6 does not satisfy its declared prediction-support gate; increased multiplicity is descriptive.
- These one-dimensional polarity observables motivate experimentally measurable local activity and heading assays. They do not establish a cell-velocity law or equate simulated heading persistence with observed migration persistence.

Proposed measurements and experimental follow-up

Common observations use native chemical 32000 and Spring 3200 horizons. Longest observations use the frozen per-condition selector, including chemical 128000 or Spring 12800 where E4 was preselected. Mixed per-size horizons remain explicit; raw total-count quotients are unavailable when exposures differ.

Persistence and its empirical bounds come from measured heading correlations with declared coverage. Bootstrap uncertainty resamples whole independent seed histories. Deterministic matched-mean/zero controls have 0 independent stochastic seeds and 0 seed-bootstrap resamples.

Primary optical counts and strict dominant-newborn takeovers use complete observed-region amounts. A takeover requires the newborn to be largest among qualified regions and its pairwise D against the previous dominant region to exceed 0.5 for two frames. With three or more regions this does not mean 75% of whole-cell activity.

Spring forcing is material attached, with physical correlation length 0.15 C/C_basal; fixed physical optical smoothing is separate. Literal A3 positions use the fixed material-origin angle times actual circumference, splitting unresolved antipodal edges; paired ±π routes are deterministic sensitivities.

Scope, limitations and pending validation

H1 is inconclusive, H2 fails the overall criterion and H3 is inconclusive; the positive selected comparison is not support for those full hypotheses.

A size advantage in heading persistence can coexist with multiple active regions. · continued

The three-model selected comparison uses 12 independent seeds per size; Goryachev remains visible in the full overview and is forced/undercovered at this input.

Late-quarter Wave-pinning/Holmes persistence ratios are bounded, not finite point estimates; unbounded bootstrap upper endpoints must remain visible.

The persistence direction and native majority-regime labels pass selected A7 checks; short-lag mobility itself is cadence-sensitive in some selected conditions and is not promoted as a robust headline.

Primary optical counts are imaging proxies at threshold 0.5, fixed physical smoothing 1.5/0.15 and cadence 80/8 in native units; native regime fractions remain distinct.

Heading persistence and local activity regions are simulated polarity observables, not measured migration persistence or a cell-velocity law.

Geometric qualifications and literal single-region site MSD remain in S4/S5; the global-heading-derived physical-mobility proxy is not one tracked site diffusivity.

Historical Spring sampled-duration/cadence failures, finite-window bounds, coverage exclusions and E4 source-recovery/original-cohort distinctions remain explicit.

Short-lag global-heading physical-mobility estimates may increase while angular persistence increases. In multiple-region states they do not measure diffusion of one tracked site. Selected mobility estimates have cadence sensitivity and are excluded from the headline.

E4 native source repair is an engineering acceptance under unchanged tolerances, not a stationarity/persistence conclusion. Retained prefixes, sigma branches, route choices, numerical companions and common/longest references are not extra independent seed replicates.

All 578 required native-atlas pages are retained in 13 numbered lossless appendix volumes. The core PDF is the statistical supplement and appendix index. Full 17-significant-digit native CSV.gz fields remain verified in the HPC archive; every page has portable exact-source inventories and compact companions.

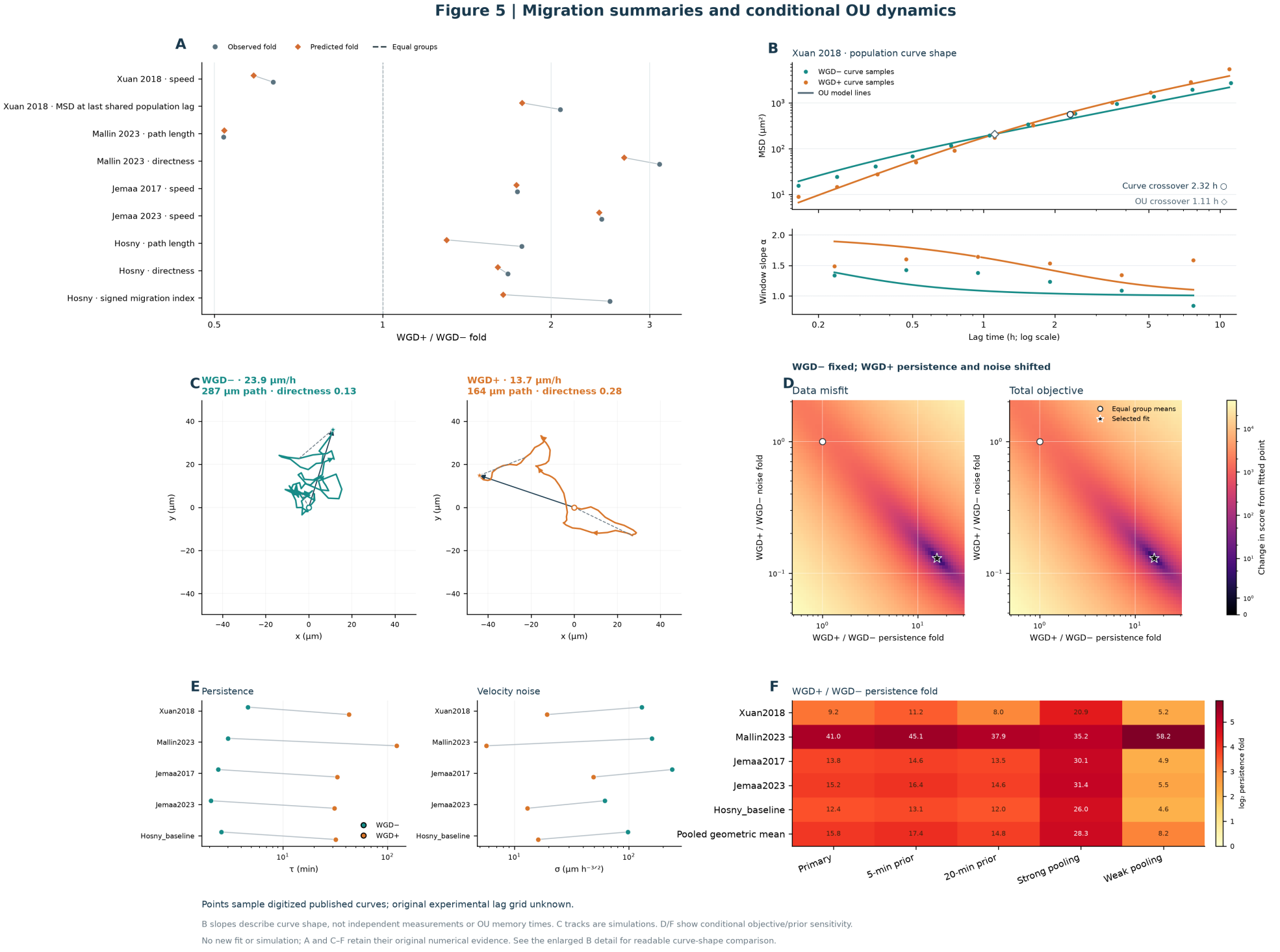
For Spring C0=3, f=1, seed202610192320800, the original full numerical gate failed 12 scalar-observation cadence comparisons. The accepted stable scope passes 180 rows and excludes all 18 q90 multiregion-span rows, including six passing ODE-only comparisons. These are consecutive saved multiregion-mask durations, not tracked individual lifetimes. The full-window q90 changes from 10 to 7.15 when scalar cadence changes from 0.5 to 0.05. Exact sigma-one replay preserves an existing path and forcing; it does not establish numerical or observation convergence. Numerical companions and dense samples are not independent scientific seeds.

A size advantage in heading persistence can coexist with multiple active regions. · continued

Sources and reproducibility

E0
E1
E2
E3
E4
A1
A2
A3
A4
A5
A6
A7
windows_and_coverage
common_horizon
long_horizon
framing
independent_final_decision_audit
final_statistics_producer
S11_full_atlas_coverage
reviewed_interpretation_memo
reviewed_interpretation_memo_display
main_display_audit
Historical Spring observation-cadence qualification
Independent Figure 4 completion review

Figure 5. Migration observations and conditional OU interpretation · image 1 of 2



Reassembled Figure 5: original A and C–F numerical evidence, revised MSD/curve-shape panel B.

Figure 5. Migration observations and conditional OU interpretation · image 2 of 2

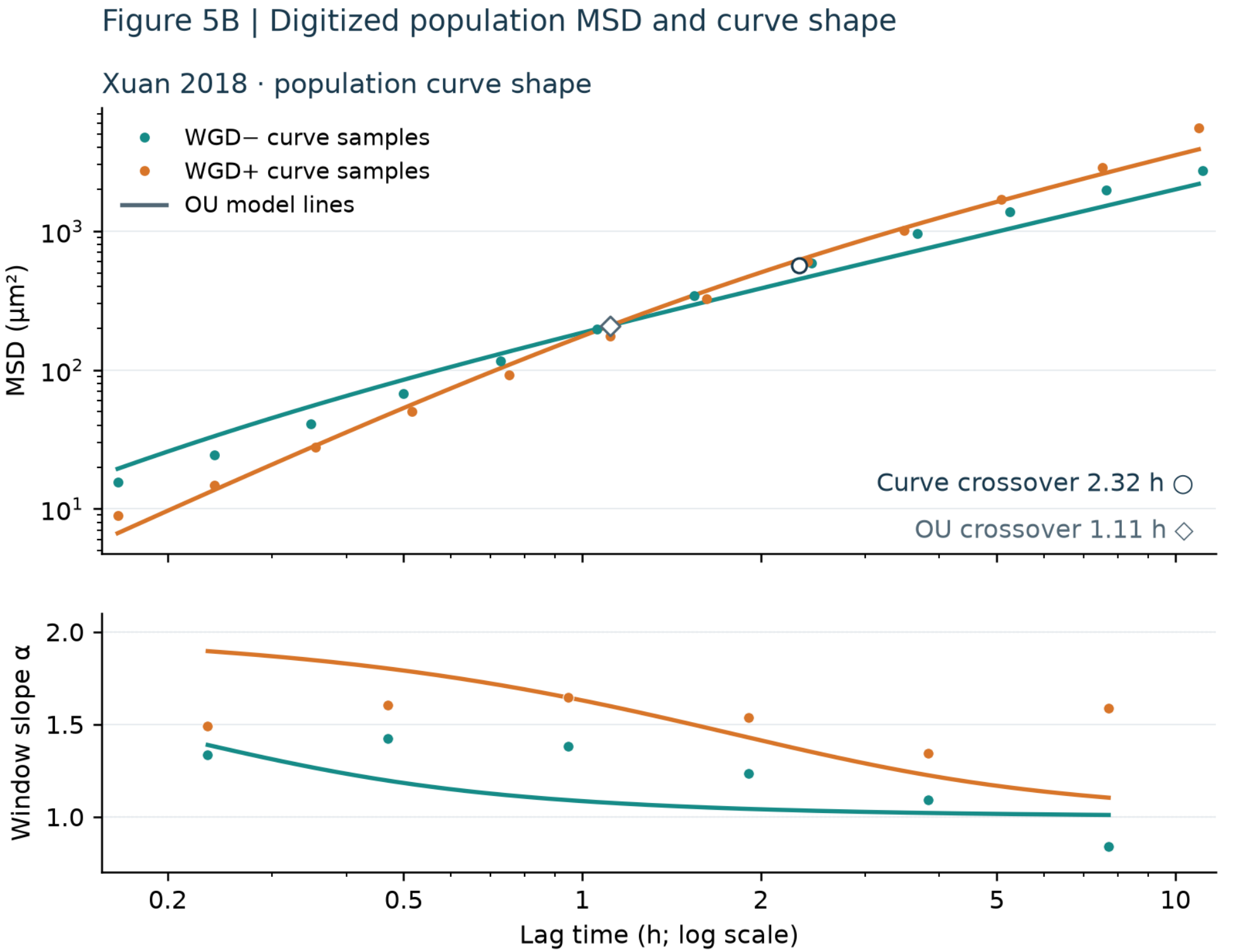


Figure 5B continued: enlarged detail with explicit raster-sample caveat.

Figure 5. Published migration summaries support a persistence hypothesis with conditional OU interpretation

Existing empirical support, with observed migration summaries kept separate from fitted velocity-memory parameters.

Detailed legend

A, Published/digitized enlarged or polyploid-group to reference-group ratios for nine migration summaries, alongside primary Ornstein–Uhlenbeck (OU) fit predictions. Xuan contributes sampled mean speed and population mean-squared displacement (MSD) at the last shared lag, approximately 10.95 h; Hosny summaries are group medians. Mallin central-bar statistic is not specified by the source and retains the historical mean approximation. These distinct source summaries are not interchangeable measures of persistence.

B, Upper: twelve chosen display samples from each digitized Xuan population MSD curve (small points) and the unchanged fitted analytical OU expectation (solid lines), on logarithmic lag and MSD axes. These are samples of the published raster curves, not recovered experimental lag positions or independent cell observations; the original experimental lag grid is unknown. The open circle identifies the crossover of the full digitized traces at approximately 2.32 h, computed by linear interpolation in native lag and MSD. The open diamond identifies the different fitted OU crossover at approximately 1.11 h. Lower: descriptive slopes of log(MSD) versus log(lag) fitted in six equal log-lag windows of the full digitized curves (points), and matching-window moving OU slopes (solid lines). Each window spans approximately 0.304 decades; both observed and model slopes use the same full window support. Observed raster columns remain correlated and give no biological sample size or confidence interval. These slopes are newly calculated curve-shape summaries, not author-reported local exponents or fitted OU memory times. A constant power-law exponent would appear horizontal; reference values 1 and 2 indicate diffusive and straight-line scaling. The model does not reproduce the complete observed curvature or the observed crossover. The analytical fit, all fitting targets and numerical inputs to A and C–F are unchanged.

C, Representative Xuan primary-fit model-generated tracks, with sampled speed, total path length and directness (net displacement divided by path length). Both groups use identical spatial scales. The tracks are simulations, not published individual-cell trajectories; a slower sampled trajectory can nevertheless have greater endpoint directness. Dashed chords illustrate the contribution of finite-time displacements.

D, With reference-group parameters fixed, vary enlarged-group persistence and velocity-noise ratios. Separate surfaces show change in the descriptive data-misfit score and in the total regularized objective; the selected fit and equal-group reference are marked. The heuristic score is not a calibrated likelihood, and the surfaces are not confidence regions.

Figure 5. Published migration summaries support a persistence hypothesis with conditional OU interpretation · continued

E, Primary paired persistence and velocity-noise estimates for Xuan 2018, Mallin 2023, Jemaà 2017, Jemaà 2023 and the Hosny hypoxic baseline. F, Enlarged/reference persistence folds under the primary 10-minute prior, 5- and 20-minute alternatives and stronger/weaker pooling, with the pooled geometric mean. Priors and pooling define conditional sensitivity scenarios, not uncertainty intervals. The Jemaà comparisons constrain speed alone, leaving persistence/noise tradeoffs dependent on hierarchy and priors.

The fitted population contrasts span stable tetraploids, giant polyploids and treatment-induced polyaneuploid cancer cells (PACCs). The inherited WGD+/WGD− labels in the artwork are comparison shorthand and do not establish matched WGD-only interventions in every study. Hosny observations derive from the recovered preprint; equivalence to the journal version remains unresolved.

Narrative points and conclusions

- Xuan combines lower sampled speed (0.64-fold) with greater late-lag population MSD (2.08-fold); Mallin combines shorter path length (0.52-fold) with greater directness (3.12-fold). Persistence-related behaviour therefore deserves analysis separately from speed.
- Hosny baseline PACCs have greater median path length (1.77-fold) and directness (1.67-fold). Signed migration index is a separate directional summary.
- All five primary OU comparisons infer longer velocity memory. The pooled primary persistence fold is about 15.8, and remains above one across the declared prior/pooling scenarios; this is a conditional model result.
- The fit does not reproduce the complete Xuan curve shape within its raster envelope, and data-only sensitivity has rank 18 of 22. Optimization completion does not establish model adequacy or uniquely identified persistence times.
- The observed Xuan curves cross near 2.32 h while the unchanged fitted OU curves cross near 1.11 h. Matched-window slopes make the remaining curve-shape discrepancy visible; fitted persistence is a conditional summary.
- Lower sampled speed can coexist with greater late-lag displacement in the enlarged-group comparison. Its causal connection to polarity mechanisms requires matched activity and migration measurements.

Proposed measurements and experimental follow-up

The new WGD assay should measure matched trajectories directly, with declared sampling and observation duration, and assess speed, displacement, directness and lag-dependent MSD before considering fitted persistence. An activity-polarity memory

Figure 5. Published migration summaries support a persistence hypothesis with conditional OU interpretation · continued

measurement would complement these quantities without equating activity and velocity.

Scope, limitations and pending validation

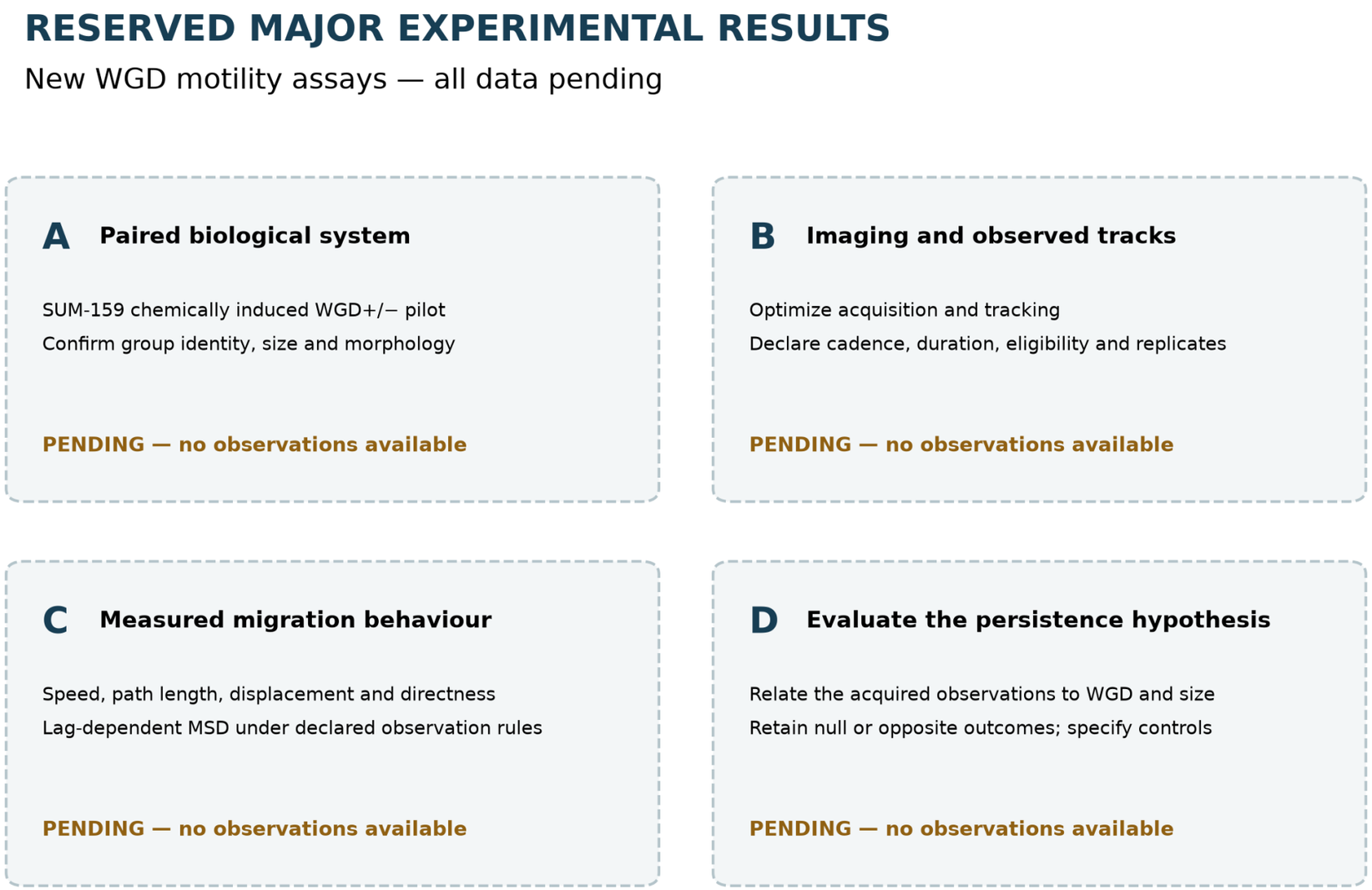
Neither these heterogeneous group comparisons nor the OU fits establish that enlargement alone causes persistence. Cell identity, induction/treatment and acquisition differences require explicit treatment in the new experiments.

The requested observed-point/model-line and curve-shape display is implemented. The original experimental lag grid remains unknown; neither display samples nor raster-derived slopes supply independent biological replicates. The observed/model crossover discrepancy remains visible.

Sources and reproducibility

Refined OU analysis
Original panel legend
Observed folds
Fit results and adequacy
Hosny sensitivity supplement
Revised display, exact preserved inputs and direct-user feedback
Migration display numerical and preservation checks

Figure 6. RESERVED: new WGD motility results



Explicit placeholder. Every panel is pending acquisition, analysis and author review; no mock observations are displayed.

Figure 6. RESERVED: new WGD motility-assay results

A substantial new experimental results section and main figure. No observations are available; the panel layout is a proposal, not an experimental result.

Detailed legend

A, Proposed experimental context: the SUM-159 chemically induced WGD+/− pair is the agreed starting pilot with the Speros collaboration. Imaging should first be optimized for that pair before expansion to additional cell lines. Ploidy/group identity, physical size and morphology need to be documented so the intended comparison is clear.

B, Reserved for representative experimental fields and trajectories under the selected acquisition conditions. The eventual panel should state sampling interval, usable duration, spatial scale, track eligibility and exclusions. Number of independent preparations and biological replicates remains to be specified.

C, Reserved for empirical motility summaries: sampled speed, accumulated path length, endpoint displacement/directness and lag-dependent MSD. The final readouts, follow-up window and analysis plan are to be settled before interpreting the data; this placeholder does not imply that all listed measurements will be technically feasible.

D, Reserved for the measured relationship between WGD-associated enlargement and migration behaviour, including size distributions and controls needed to interpret the group contrast. Any activity/protrusion measurements arising from the mechanism proposals require a separately agreed assay and should be distinguished from the basic motility pilot. Statistical models, uncertainty reporting and sample-size targets remain open.

Narrative points and conclusions

- This major results section will test whether the planned paired experimental system exhibits the migration behaviour motivating the theoretical work.
- Its conclusions must follow the acquired observations, including null or opposing results. Simulation output and literature fits cannot fill this gap.
- Agreement between increased polarity resistance or activity memory and migration persistence would support the proposed connection; causal attribution requires appropriate perturbations and controls.

Proposed measurements and experimental follow-up

Pending decisions include plating density, acquisition duration and cadence, field/condition throughput, tracking and exclusion rules, replicate structure, confirmation of WGD status, size quantification and the statistical analysis. The known throughput versus spatial/temporal resolution trade-off should be resolved by the pilot.

Figure 6. RESERVED: new WGD motility-assay results · continued

Mechanism validation can subsequently examine incumbent extent, early challenger growth, local loss/rebound, site fates and global activity-polarity correlation under characterized perturbations. These activity experiments are proposals alongside the theoretical mechanisms and are separate from the reserved motility-results contribution.

Scope, limitations and pending validation

No pilot acquisition, sample size, effect estimate, significance claim or experimental outcome is supplied by the available evidence.

Sources and reproducibility

Author manuscript vision and supplied collaboration context

Supplementary material and supporting analyses

Figure 2: complete revised evidence

The comprehensive supplement retains the full designed condition grid, unavailable measurements, finite-horizon bounds, native histories, controls and the stated numerical and observational sensitivities. Earlier figure versions remain development history.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/signature_assays_20261020/supporting/vectors/figure2_signature_assays_complete_supplement.pdf

Figure reader and source data

Independent completion review

Figure 3: complete revised evidence

The comprehensive supplement retains the full designed condition grid, unavailable measurements, finite-horizon bounds, native histories, controls and the stated numerical and observational sensitivities. Earlier figure versions remain development history.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261020/figure3_distance/supporting/vectors/figure3_distance_supplement.pdf

Figure reader and source data

Independent completion review

Figure 4: complete revised evidence

The comprehensive supplement retains the full designed condition grid, unavailable measurements, finite-horizon bounds, native histories, controls and the stated numerical and observational sensitivities. Earlier figure versions remain development history.
S0–S12 statistical core and complete native-atlas appendix index; native pages are in all linked appendix volumes. Every numbered appendix below is part of the complete supplement.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/signature_figure4_20261020/supporting/vectors/figure4_revision_complete_supplement.pdf

Figure reader and source data

Independent completion review

S11 native atlas appendix 001

S11 native atlas appendix 002

S11 native atlas appendix 003

S11 native atlas appendix 004

S11 native atlas appendix 005

S11 native atlas appendix 006

S11 native atlas appendix 007

S11 native atlas appendix 008

S11 native atlas appendix 009

S11 native atlas appendix 010

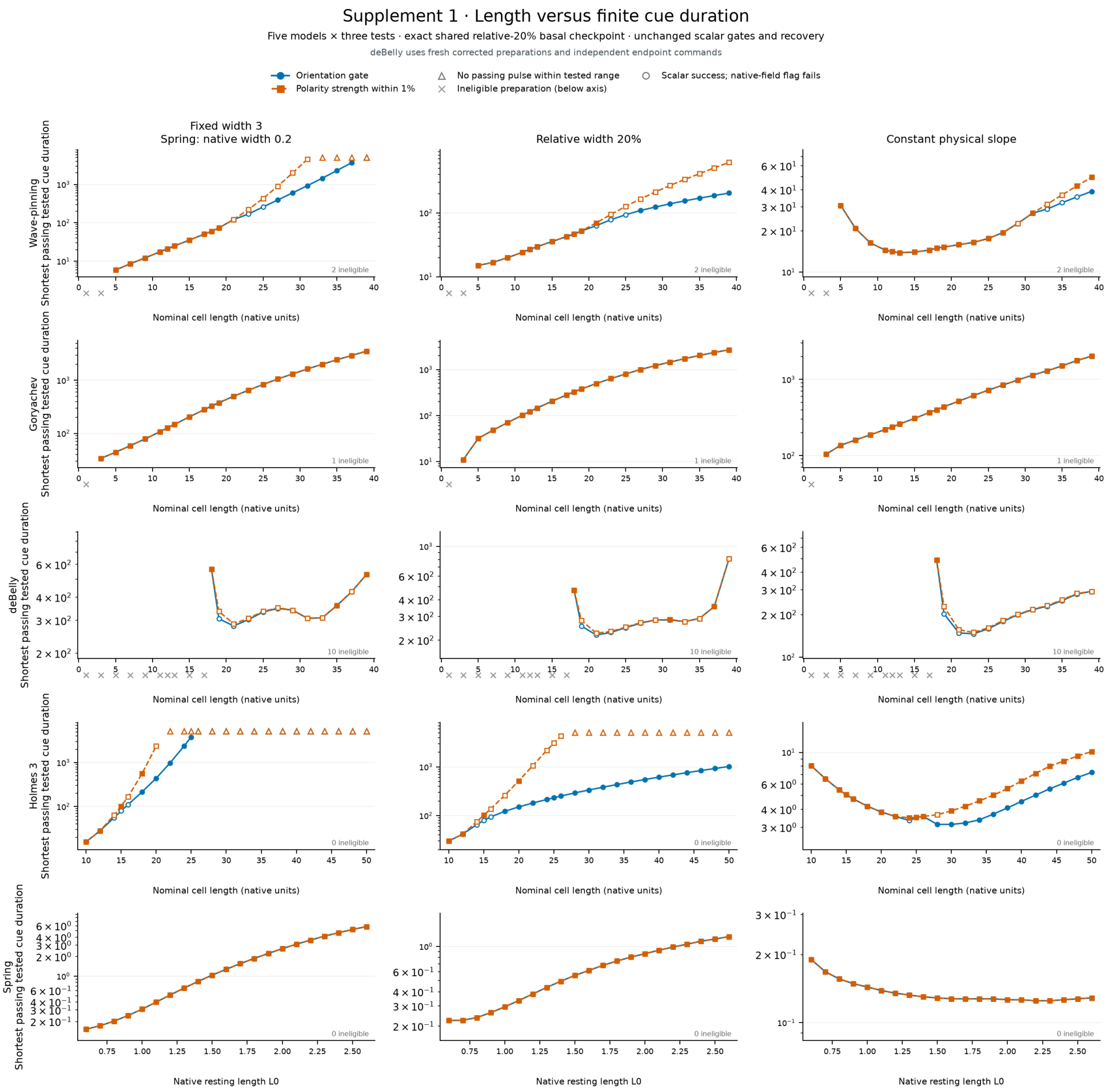
S11 native atlas appendix 011

Supplementary material and supporting analyses · continued

S11 native atlas appendix 012

S11 native atlas appendix 013

S1. Full cue-duration response curves



Both patches use a constant peak across all lengths; Goryachev K=64, peak=6.4. Gradients retain one constant physical slope per model.
Each positive duration restarts the original checkpoint. Full strength adds the 1% reversed initial-polarity band to the orientation gate.
Open markers retain failing field-tail flags. Censoring marks explicit tested limits; missing endpoints break lines. Native times and scales differ.
Data generated 2026-10-08T19:26:33.200718+00:00; producer finished. Every imported and fresh trial passed independent native-array audit.

Canonical corrected five-model duration curves; source working number Supplement 1.

Earlier supporting material: S1. Full cue-duration response curves

Essential curve context for the main comparison heatmap.

Detailed legend

Five model rows and three finite-cue columns retain the complete sampled length dependence. Coloured curves distinguish orientation from strength replacement. Open success markers retain scalar successes that fail the separate native-field tail flag; triangles indicate no passing pulse within the tested range; crosses indicate ineligible preparations. Missing values are not zero response times. Each duration excludes preparation and the subsequent 10,000-unit basal recovery. Spring uses native length and time. See Figure 1 for exact gates and stimulus conventions.

Narrative points and conclusions

- Corrected deBelly is nonmonotone across all three geometries. The positive rank coefficients summarize association rather than its full curve shape.
- Constant-slope Spring shows the opposite association to its patch responses. Holmes orientation under that cue varies with little monotone association. Censoring and native-field flags restrict interpretation even where rank correlation is strong.

Sources and reproducibility

Current figure legend and gate definitions

Supplementary material and supporting analyses

Earlier supporting material: S2. Numerical qualification and preparation provenance

Exact common starting states, duration-boundary audits, corrected deBelly endpoint controls and selected mesh diagnostics; source histories remain distinguishable from compact committed summaries.

polarity1d/manuscript/THREE_TESTS.md

Earlier supporting material: S3. Mechanism and stochastic sensitivity

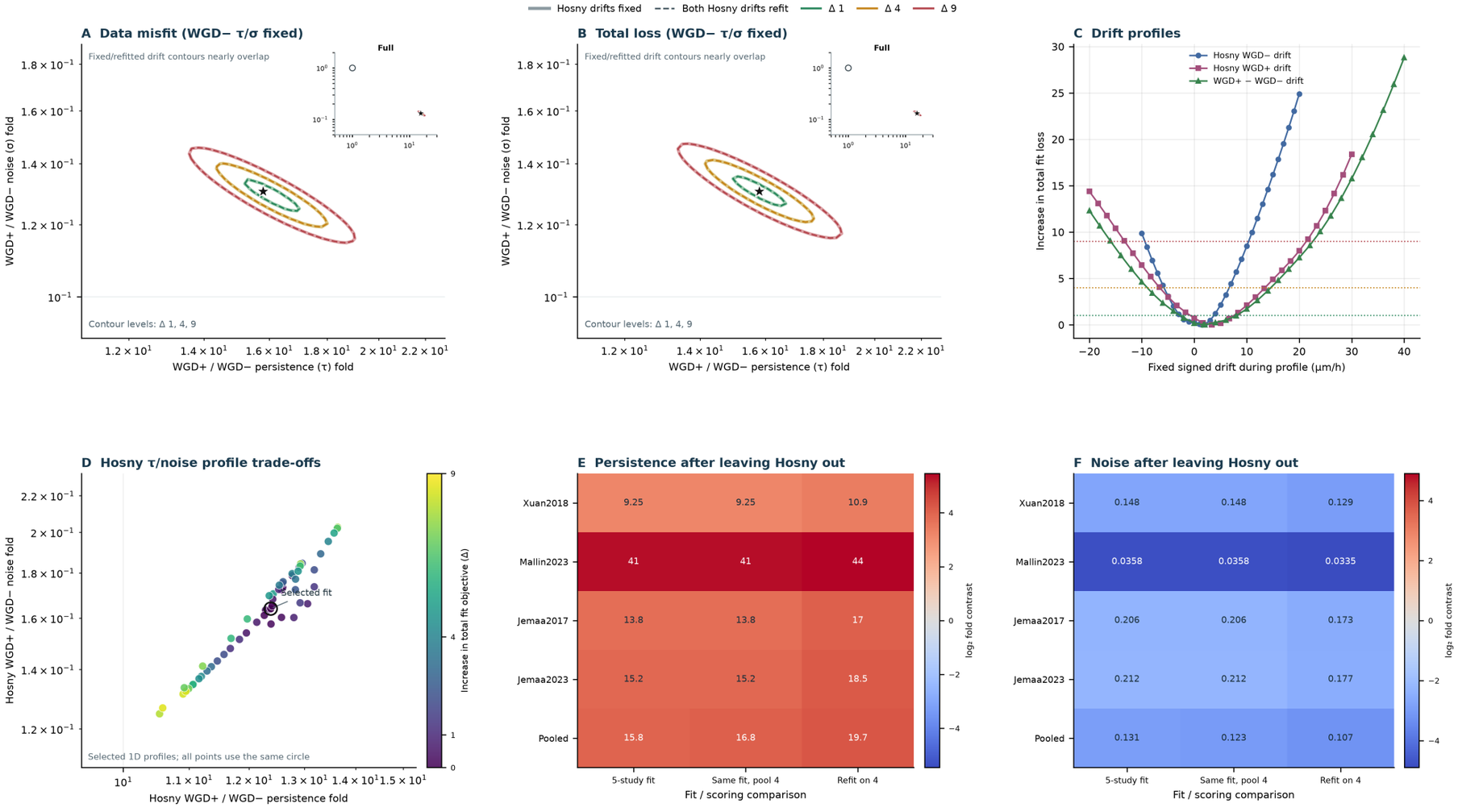
Complete gain/recovery grids, clock qualification, finite-cue definitions, outcome tables, site/vector diagnostics and numerical/noise controls accompany the mechanism checkpoint. Revised Figure 4 also displays both strengths, complete half-window diagnostics, physical-site eligibility and matched C20/C60 native examples; compact source tables support portable rerendering.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/VISION_CHECKPOINT_REQUIREMENTS.md

S4. Migration-model adequacy and Hosny sensitivity

How much does the Hosny drift assumption change the fitted contrast?

Primary Gaussian OU hierarchy · profile panels reoptimize remaining parameters · Δ values are descriptive fit sensitivity



Objective is arbitrarily weighted and has no calibrated statistical or biological units; it is not a likelihood. Profile ranges are unions of sampled 1D fits, not joint uncertainty regions. Hosny contributes medians to fitting.

Existing Hosny sensitivity supplement from the refined OU analysis.

Earlier supporting material: S4. Migration-model adequacy and Hosny sensitivity

Supporting conditional sensitivity analysis; it is not an independent biological replication.

Detailed legend

The retained supplement explores signed-drift and leave-Hosny-out sensitivity within the declared OU hierarchy. Its fitted changes and descriptive-score profiles are conditional on the model and supplied summaries, not confidence intervals. Data-only identifiability, Xuan population-curve residuals and prior/pooling assumptions remain part of the adequacy assessment. Representative movie diagnostics are separate: acquisition timing is uncertain and nuclear appearance does not establish ploidy.

Sources and reproducibility

Original Hosny supplement caption

Sensitivity methods and results

Earlier supporting material: Gain and small-loss recovery: supporting evidence

Detailed legend

Calculated native input gain is separate from observed post-cue activity. The small-loss recovery contrast persists through stored finite trajectories but weakens with time; actual damage is only about0.22–0.285% of core activity, with no reporter calibration.

Scope, limitations and pending validation

Small-loss assays are not the finite optical challenge or a universal stability measure.

Sources and reproducibility

Finite-time and amplitude audit

All97 gain/recovery values

Earlier supporting material: Selected native pulse and causal controls

Detailed legend

WP L15 T20/T60 starts from the same basal state and illustrates withdrawal-dependent domain survival. Evolved interventions have endpoint- and cue-specific limits.

Scope, limitations and pending validation

WP4× mobility passes orientation only at one dose; both conditions fail the stronger full-polarity-strength gate. Mobility is changed during cue and recovery.

Sources and reproducibility

Retained WP short/long pulse artwork

Endpoint-qualified controls

Supplementary material and supporting analyses

Earlier supporting material: Stationary-cue endpoint-vector alternative

An alternative view displays all 240 measured endpoint vectors on equal coordinate scales, with weak-resultant, overlap and finite-tail flags retained. It is available for discussion without replacing the complete withdrawal/retention map in the current review.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure3/standard_angles_sizes/vector_grid/README.md

Earlier supporting material: Spring: every native medoid

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_all_histories.png

Earlier supporting material: Spring magnitude, geometry and controls

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_magnitude_geometry_controls.png

Earlier supporting material: Spring full/half-window memory

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_memory_windows.png

Earlier supporting material: Spring increment tails

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_increment_tails.png

Earlier supporting material: Spring short/long-lag geometry and circular mixing

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_endpoint_lag_sensitivity.png

Earlier supporting material: Spring largest sampled increment native context

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_large_increment_context.png

Earlier supporting material: All chemical models including Goryachev

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_stochastic_revision.png

Earlier supporting material: Chemical memory: both strengths

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_both_strengths_memory.png

Supplementary material and supporting analyses · continued

Earlier supporting material: Chemical magnitude and controls

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_magnitude_controls.png

Earlier supporting material: Chemical increment tails

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_increment_tails.png

Earlier supporting material: Chemical site-gate sensitivity

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_site_gate_sensitivity.png

Earlier supporting material: Original chemical medoids, all18 conditions

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/vision_noise_20261017/vision_noise_selected_histories.png

Earlier supporting material: Goryachev lowest-resultant diagnostic

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/vision_noise_20261017/vision_noise_lowest_resultant_diagnostic.png

Earlier supporting material: Spring active amount and spatial first harmonic

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_amount_spatial_balance.png

Earlier supporting material: Preserved complete mechanism and assay checkpoint

The preceding outline retains all-five mechanism-to-measurement sheets, ring gain/recovery and finite-cue maps, and noise magnitude, controls and full/first/last-window diagnostics. Its numerical evidence is preserved; the revised main-figure claims are stated in this edition.

manuscript_outline/outline_v2.pdf

Prior all-five mechanism evidence crosswalk

Prior complete ring and noise checkpoint

Prior physical motion, magnitude and coverage evidence

References and evidence provenance

Reviewed primary literature and extraction map
Scientific scope and author direction
Direct-user preference evidence and foundational figure audit
Mechanism checkpoint intake and live-producer evidence
Source checkout, evidence selection and reproduction
Independent critical assessment and repair decisions
Portable original/renumbered mechanism replay and source closure
Second-edition integration decisions and validation scope
Figures 2–4 revision specification
Figure 2 October 20 revision brief
Figure 3 October 20 revision brief
Figure 4 October 20 revision brief